

Yuzhen Song

Southern University of Science and Technology

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 Personal Page •  GitHub Repository

Research Interests: Humanoid Locomotion and Whole-Body Control; Reinforcement Learning; Sim-to-Real Transfer.

EDUCATION

B.Eng. in Robotics Engineering

Southern University of Science and Technology

Aug 2023 – Jun 2027

Shenzhen, China

- **Overall GPA:** 3.87/4.0 (Weighted Avg: 93.03/100) **Major GPA:** 3.94/4.0
- **Core Courses:** Robot Operating System (100), Mechanics of Materials M (100), Mechatronic Systems (98), Robot Modeling and Control (97), Mechanical Design (97), Finite Element Theory (97), Intro to C Programming (97), Signals and Systems (95), Sensing Technology (95), Linear Algebra (94), Machine Learning for Eng. (93), Robotic Actuation System (93), Computer Vision (90), Fundamentals of Control Eng. (90)
- **Leadership:** President of Robotics Club & Student Organizations Council (SOC)

Visiting Student

University of California, Berkeley

Summer Sessions (Session E); Berkeley Global Access (BGA) Program (Fall)

Jul 2026 – Dec 2026

Berkeley, CA, USA

PUBLICATIONS

- J. Cao*, Y. Chen*, **Y. Song**, M. Tomizuka, C. Li, T. Tian, "TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation," *Conference on Robot Learning (CoRL)*, 2026. **Accepted.**
Also presented at the WCBM and RoboData workshops at RSS 2026.

RESEARCH EXPERIENCES

University of California, Berkeley

Visiting Undergraduate Researcher, Advisor: Prof. Masayoshi Tomizuka

Berkeley, CA, USA

Apr 2026 – Sept 2026

- **Project: TEXEDO – Test-Time Scaling for Controller-Aware Humanoid Motion Generation**
 - ▷ Integrated Kimodo motions into MuJoCo via mjlab, aligning quaternion conventions, sampling rates, and evaluation logs.
 - ▷ Compared TEXEDO with Kimodo using MPJPE, velocity errors, and acceleration errors; resolved memory failures through streaming computation and logging.
 - ▷ Built a Gemini video evaluator for semantic alignment and physical plausibility, comparing automated scores with manual ratings.
 - ▷ Co-authored TEXEDO, accepted to **CoRL 2026** and presented at the RSS 2026 WCBM and RoboData workshops.

Southern University of Science and Technology

Undergraduate Researcher, Advisor: Prof. Chenglong Fu

Shenzhen, China

Sep 2024 – Present

- **Project: A Model for Predicting Metabolically Optimal Anteroposterior Center-of-Mass Assistance During Uphill Walking**
 - ▷ Reproduced the team's equations for single and double support and implemented code to predict and plot optimal assistance across slopes.
 - ▷ Designed a cable assistance platform with a motorized spool, a rail and pulley mechanism for angle adjustment, and a waist belt attachment.
 - ▷ Participating in ongoing treadmill experiments to evaluate metabolic responses to predicted assistance.

Southern University of Science and Technology

Undergraduate Researcher, Advisors: Prof. Jiansheng Dai, Prof. Zhao Tang

Shenzhen, China

Jun 2025 – Oct 2025

- **Project: Simulation of an Underactuated Metamorphic Palm for a Three-Fingered Hand**
 - ▷ Built Isaac Sim and Webots environments for an underactuated palm with five links, adding closure constraints to URDF models for preliminary RL experiments.
 - ▷ Simplified meshes while preserving geometry at contact regions, resolving GPU memory failures and enabling simulation on an 8GB GPU.
 - ▷ Investigated instability during palm reconfiguration by analyzing joint constraints and tuning damping, stiffness, and friction, identifying limitations in simulation fidelity.

SELECTED PROJECTS

Southern University of Science and Technology

Project Leader, Advisor: Prof. Chengzhi Hu

Shenzhen, China

Mar 2026 – Jun 2026

- **Project: BEAT – Barometric Elevation and Attitude Tracking for Posture-Independent Wrist Blood Pressure Measurement**
 - ▷ Led BEAT development, using barometric sensing to correct for wrist height relative to the heart.
 - ▷ Programmed two ESP32 boards for ESP-NOW communication and built a desktop UI for live sensor monitoring.
 - ▷ Designed and analyzed pilot tests across three arm positions ($n = 2$); compensation reduced systolic MAE by 56.2%.

Core Team Member, Advisor: Prof. Feng Zheng

Mar 2026 – Jun 2026

- **Project: SplatSLAM – Indoor Reconstruction with SLAM and 3D Gaussian Splatting**
 - ▷ Integrated MAST3R-SLAM with Nerfstudio's 3DGS pipeline to generate point clouds, rendered images, and walkthrough videos from monocular video.
 - ▷ Analyzed how training iterations affected PSNR, L1 loss, and visual quality, identifying signs of overfitting and spiky rendering artifacts.

Project Leader, Course Advisor: Prof. Wende Ke

Sep 2025 – Dec 2025

- **Project: ROS 2 Mobile Manipulator for Navigation and Grasping**
 - ▷ Integrated slam_toolbox, Nav2, and EKF fusion of wheel odometry and IMU data for mapping, localization, and navigation in Gazebo.
 - ▷ Integrated OpenCV visual alignment and arm control for object retrieval, with voice and gesture interfaces using Vosk and MediaPipe.

Project Leader, Course Advisor: Prof. Chenglong Fu

Mar 2025 – Jun 2025

- **Project: Image Processing and Motion Planning for Robotic Drawing**
 - ▷ Developed OpenCV routines for contour extraction and skeletonization to generate drawing paths for a six-axis robotic arm.
 - ▷ Used inverse kinematics in MATLAB to convert planar drawing paths into joint trajectories and verified the resulting motion in simulation.
 - ▷ Implemented tool orientation adjustment along surface normals for simulated drawing on cylindrical surfaces.

Project Leader, Advisors: Prof. Dong Lu, Yuanjian Huang

Dec 2024 – Aug 2025

- **Project: Autonomous Rescue Robot (National Competition)**
 - ▷ Led the development of a mobile robot integrating YOLOv11 object detection and STM32 motion control for autonomous target tracking and retrieval.
 - ▷ Won **National Second Prize** and **Provincial First Prize** in the China Undergraduate Engineering Practice and Innovation Ability Competition.

SKILLS

- **Programming:** C/C++, Python, MATLAB
- **Robotics & Simulation:** ROS 2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, Webots
- **Computer Vision & Visualization:** OpenCV, RViz2
- **Embedded Systems:** STM32, ESP32
- **CAD/CAM & CAE:** SolidWorks, AutoCAD, Edgcam, Abaqus

HONORS & AWARDS

- National Second Prize, China Undergraduate Eng. Practice and Innovation Ability Competition 2025
- Provincial First Prize, Guangdong Undergraduate Eng. Practice and Innovation Ability Competition 2025
- First-Class Scholarship for Outstanding Students (Top 3%) 2025
- Provincial Second Prize, Guangdong Undergraduate Mechanical Innovational Competition 2025
- Outstanding Student Award 2024, 2025, 2026
- Second-Class Scholarship for Outstanding Students (Top 8%) 2024, 2026